THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Ethanol is miscible with water because: 1

(a) it is heavy **(b)** it is acid

(c) it is non-polar **(d)** it can form hydrogen bonds

2. The molecular formula of an alkyne with 4 carbon atoms is: 1

(a) C_4H_8 **(b)** C_4H_6 **(c)** C_4H_{10} **(d)** C_4H_4

3. A compound X reacts with Na to give H_2 and with sodium carbonate gives CO_2. X is: 1

(a) ethanal **(b)** ethane **(c)** ethanol **(d)** ethanoic acid

4. Compare boiling points: methane, ethanol, water. The highest is of: 1

(a) methane **(b)** equal **(c)** water **(d)** ethanol

5. The IUPAC name of $CH_3CH_2CH_2COOH$ is: 1

(a) butanal **(b)** propanoic acid **(c)** butanol **(d)** butanoic acid

6. A compound that reacts with sodium but not with sodium carbonate is: 1

(a) ethanol **(b)** ethene **(c)** methanoic acid **(d)** ethanoic acid

7. The next homologue of CH_3COOH is: 1

(a) CH_3CH_2OH **(b)** $HCOOH$ **(c)** C_2H_5COOH **(d)** CH_3COOCH_3

8. The IUPAC name of CH_3COCH_3 is: 1

(a) propanone **(b)** propanal **(c)** propanol **(d)** propanoic acid

9. Unsaturated hydrocarbons are converted to saturated by: 1

(a) saponification **(b)** esterification **(c)** hydrogenation **(d)** oxidation

10. The number of isomers for C_4H_{10} is: 1

(a) 1 **(b)** 2 **(c)** 4 **(d)** 3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the molecular formula and name of the compound that forms by oxidation of propanol. 2

12. Why does an alkane not undergo addition reactions? 2

13. What is the role of conc. H_2SO_4 in esterification? 2

14. Name the products of burning ethanol and write the balanced equation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Compound X (C2H4O2) reacts with NaHCO3 to give a gas. Identify X, the gas and write the equation. 3

16. Why are the boiling points of alcohols higher than hydrocarbons of similar mass? 3

17. Why do some carbon compounds like CCl_4 not burn? 3

OR

Why is the combustion of hydrocarbons in a limited air supply harmful? Name the gas. 3

18. Name the functional group and write the next homologue: 3
(a) CH_3CHO
(b) C_2H_5OH.

OR

How is ethanoic acid distinguished from ethanol using litmus? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Compound A (C_2H_6O) on oxidation gives B ($C_2H_4O_2$). A with B in presence of conc. H_2SO_4 gives C. Identify A, B and C and write the equations. 5

OR

Describe the uses of carbon compounds: fuels, soaps, solvents, medicines, plastics with one example each. 5

20. Explain how the structure of diamond, graphite and fullerene explains conductivity and hardness. 5

OR

Describe the covalent compounds' physical properties (mp, conductivity, solubility) with reasons. 5

END OF QUESTION PAPER — Total: 40 Marks

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